

Problem

Refer to the oscillator in Fig. 10.137.

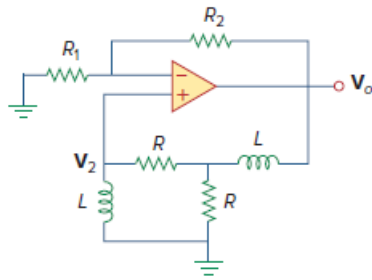
(a) Show that

$$\frac{V_2}{V_o} = \frac{1}{3 + j(\omega L/R - R/\omega L)}$$

(b) Determine the oscillation frequency f_o .

(c) Obtain the relationship between R_1 and R_2 in order for oscillation to occur.

Figure 10.137:



Step-by-step solution

Step 1 of 8

Draw the following circuit diagram:

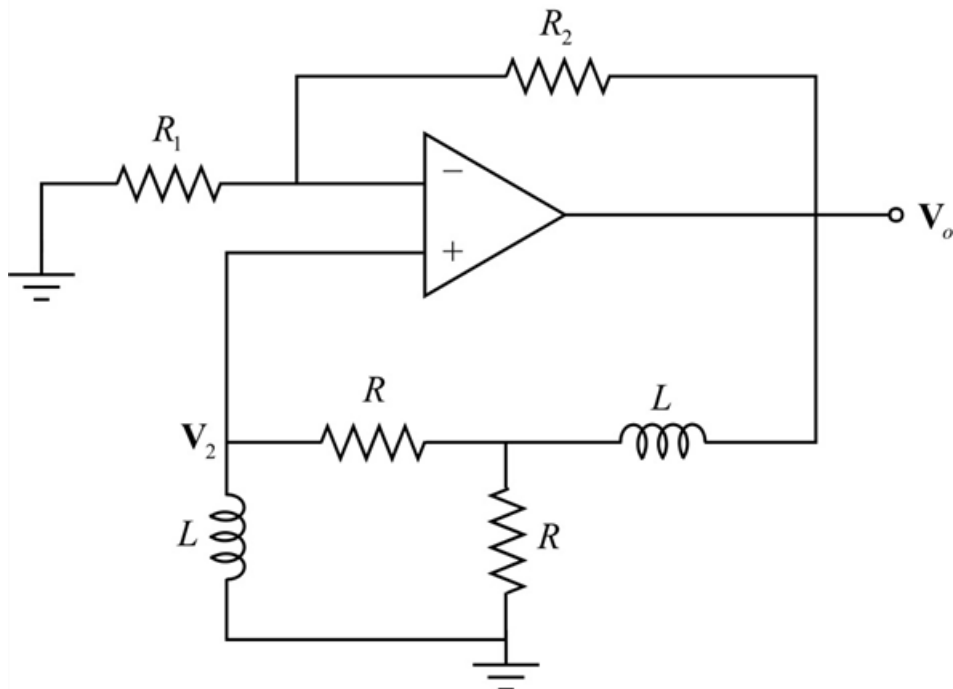


Figure 1

Step 2 of 8

(a)

Draw the feedback part of the circuit.

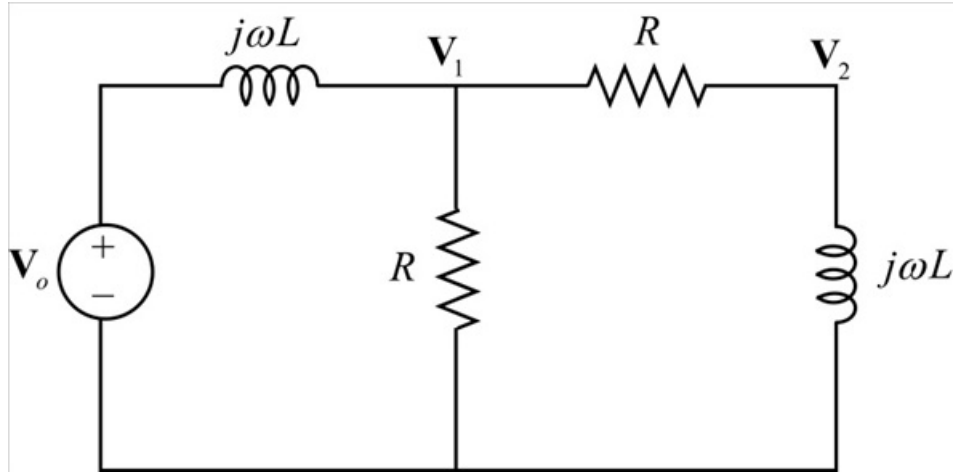


Figure 2

Step 3 of 8

Apply voltage division rule for the circuit in Figure 2.

$$\mathbf{V}_2 = \left[\frac{j\omega L}{R + j\omega L} \right] \mathbf{V}_1$$

$$\mathbf{V}_1 = \left[\frac{R + j\omega L}{j\omega L} \right] \mathbf{V}_2$$

Apply Kirchhoff's current law at the node $\mathbf{V}_1$ for the circuit in Figure 2.

$$\frac{\mathbf{V}_1 - \mathbf{V}_o}{j\omega L} + \frac{\mathbf{V}_1}{R} + \frac{\mathbf{V}_1}{R + j\omega L} = 0$$

$$\mathbf{V}_1 \left[\frac{1}{j\omega L} + \frac{1}{R + j\omega L} + \frac{1}{R} \right] = \frac{\mathbf{V}_o}{j\omega L}$$

$$\mathbf{V}_o = \mathbf{V}_1 \left[\frac{1}{j\omega L} + \frac{1}{R + j\omega L} + \frac{1}{R} \right] (j\omega L)$$

Step 4 of 8

Substitute $\left[\frac{R + j\omega L}{j\omega L} \right] \mathbf{V}_2$ for $\mathbf{V}_1$ in $\mathbf{V}_o = \mathbf{V}_1 \left[\frac{1}{j\omega L} + \frac{1}{R + j\omega L} + \frac{1}{R} \right] (j\omega L)$.

$$\mathbf{V}_o = \left[\frac{R + j\omega L}{j\omega L} \right] \left[\frac{1}{j\omega L} + \frac{1}{R + j\omega L} + \frac{1}{R} \right] (j\omega L) \mathbf{V}_2$$

$$= [R + j\omega L] \left[\frac{1}{j\omega L} + \frac{1}{R + j\omega L} + \frac{1}{R} \right] \mathbf{V}_2$$

$$= \left[\frac{R + j\omega L}{j\omega L} + 1 + \frac{R + j\omega L}{R} \right] \mathbf{V}_2$$

$$= \left[\frac{R(R + j\omega L) + j\omega LR + j\omega L(R + j\omega L)}{j\omega LR} \right] \mathbf{V}_2$$

Step 5 of 8

Simplified V_o further:

$$V_o = \left[\frac{(R + j\omega L)(R + j\omega L) + j\omega LR}{j\omega LR} \right] V_2$$

$$V_o = \left[\frac{R^2 + 2j\omega LR - \omega^2 L^2 + j\omega LR}{j\omega LR} \right] V_2$$

$$V_o = \left[\frac{R^2 + 3j\omega LR - \omega^2 L^2}{j\omega LR} \right] V_2$$

$$V_2 = \left[\frac{j\omega LR}{R^2 + 3j\omega LR - \omega^2 L^2} \right] V_o$$

Calculate the function $\frac{V_2}{V_o}$.

$$\frac{V_2}{V_o} = \frac{j\omega LR}{j\omega LR \left(3 + \frac{R^2 - \omega^2 L^2}{j\omega LR} \right)}$$

$$= \frac{1}{3 + \frac{(R^2 - \omega^2 L^2)}{j\omega LR}}$$

$$= \frac{1}{3 + j \left(\frac{\omega^2 L^2}{\omega LR} - \frac{R^2}{\omega LR} \right)}$$

$$= \frac{1}{3 + j \left(\frac{\omega L}{R} - \frac{R}{\omega L} \right)}$$

Therefore, the function $\frac{V_2}{V_o} = \frac{1}{3 + j \left(\frac{\omega L}{R} - \frac{R}{\omega L} \right)}$ is proved.

Step 6 of 8

(b)

In order to keep the gain $\frac{V_2}{V_o}$ real and to obtain oscillations, the function should not contain imaginary part.

$$\frac{\omega_o L}{R} - \frac{R}{\omega_o L} = 0$$

$$\frac{\omega_o L}{R} = \frac{R}{\omega_o L}$$

$$(\omega_o L)^2 = R^2$$

$$\omega_o L = R$$

Substitute $2\pi f_o$ for ω_o in $\omega_o L = R$.

$$2\pi f_o L = R$$

$$f_o = \frac{R}{2\pi L}$$

Therefore, the oscillation frequency f_o is $\boxed{\frac{R}{2\pi L}}$.

Step 7 of 8

(c)

In order to keep the gain $\frac{V_2}{V_o}$ real and to obtain oscillations, the function should not contain imaginary part.

$$\frac{\omega_o L}{R} - \frac{R}{\omega_o L} = 0$$

Substitute 0 for $\frac{\omega_o L}{R} - \frac{R}{\omega_o L}$ in $\frac{V_2}{V_o} = \frac{1}{3 + j\left(\frac{\omega L}{R} - \frac{R}{\omega L}\right)}$.

$$\frac{V_2}{V_o} = \frac{1}{3 + j(0)}$$

$$\frac{V_2}{V_o} = \frac{1}{3}$$

$$\frac{V_o}{V_2} = 3 \quad \left(\text{since } A_v = \frac{V_o}{V_2} \right)$$

$$A_v = 3$$

Step 8 of 8

Calculate the gain of the amplifier:

$$A_v = 1 + \frac{R_2}{R_1}$$

Substitute 3 for A_v in $A_v = 1 + \frac{R_2}{R_1}$.

$$3 = 1 + \frac{R_2}{R_1}$$

$$\frac{R_2}{R_1} = 2$$

$$R_2 = 2R_1$$

Therefore, the relationship between R_1 and R_2 in order for oscillation to occur is $R_2 = 2R_1$.